

PORT MEADVILLE, PA, USA

WMO: 725104

Lat: 41.626N

Lon: 80.215W

Elev: 1400

StdP: 13.97

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04843

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.1	6.9	-5.8	4.2	5.4	-1.1	5.5	8.8	22.7	31.3	20.1	27.4	7.4	210	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	85.7	71.5	83.3	70.4	81.2	69.4	74.7	81.9	72.9	79.7	71.4	77.8	7.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	126.2	79.0	70.8	119.4	76.8	69.6	114.7	75.5	39.0	82.4	37.4	79.5	36.0	78.5	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.7	14.7	DB	-2.4	89.2	6.2	2.8	-6.9	91.2	-10.6	92.8	-14.1	94.3	-18.6	96.4	(4)
(5)			WB	-2.9	77.8	6.1	2.9	-7.2	79.9	-10.8	81.6	-14.2	83.2	-18.7	85.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.1	25.2	27.2	35.4	48.1	58.8	66.4	70.7	69.1	63.2	51.7	41.1	30.8	(6)
(7)		DBStd	18.41	11.72	11.20	11.54	9.95	8.68	6.57	5.21	4.95	7.03	8.78	9.46	9.77	(7)
(8)		HDD50	3040	770	643	475	151	21	0	0	0	2	86	294	598	(8)
(9)		HDD65	6357	1232	1060	918	512	231	62	11	19	115	421	716	1060	(9)
(10)		CDD50	2714	3	4	23	95	295	492	642	591	399	139	28	3	(10)
(11)		CDD65	551	0	0	0	5	39	104	188	144	62	9	0	0	(11)
(12)		CDH74	3460	0	0	4	47	282	676	1233	807	375	36	0	0	(12)
(13)		CDH80	719	0	0	0	3	42	139	301	153	78	3	0	0	(13)
(14)	Wind	WSAvg	6.6	8.2	7.8	7.5	7.6	6.3	5.7	5.2	4.9	5.1	6.4	7.3	7.9	(14)
(15)	Precipitation	PrecAvg	44.2	3.0	2.5	3.2	3.7	3.8	4.4	4.4	3.9	4.4	3.9	3.5	3.5	(15)
(16)		PrecMax	59.5	5.8	5.7	6.2	8.8	7.3	8.5	9.7	7.3	8.1	7.0	5.0	7.1	(16)
(17)		PrecMin	28.5	1.4	1.0	1.3	1.8	1.3	0.9	0.9	1.9	1.4	2.0	0.9	1.4	(17)
(18)		PrecStd	6.9	1.1	1.1	1.2	1.4	1.6	1.8	2.3	1.3	1.8	1.2	1.1	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.8	62.5	71.6	79.4	84.6	86.4	89.6	87.9	86.3	78.6	68.7	60.5	(19)
(20)			MCWB	53.3	53.7	57.9	62.5	68.6	71.0	73.5	74.3	71.0	67.0	57.2	54.8	(20)
(21)		2%	DB	53.9	54.4	63.9	74.0	80.6	83.8	86.0	83.8	82.1	73.2	63.4	54.2	(21)
(22)			MCWB	51.3	49.6	53.9	59.1	66.5	70.4	72.4	71.3	69.4	63.0	55.3	50.1	(22)
(23)		5%	DB	47.5	48.5	58.6	69.7	77.2	81.3	83.3	81.3	78.6	69.6	59.5	49.8	(23)
(24)			MCWB	43.8	43.8	50.5	56.7	64.5	70.0	70.9	69.7	67.6	61.8	52.2	46.2	(24)
(25)		10%	DB	41.5	43.1	53.7	64.2	73.2	78.3	81.0	78.9	74.7	65.3	55.4	45.0	(25)
(26)			MCWB	38.0	37.9	47.0	53.6	62.0	67.0	69.8	68.0	65.7	59.0	49.2	42.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.7	55.9	59.6	66.1	71.9	75.5	77.6	75.8	75.0	70.0	60.7	55.5	(27)
(28)			MCDWB	57.4	60.7	69.1	74.7	80.0	80.9	86.6	84.0	82.3	76.1	65.8	58.5	(28)
(29)		2%	WB	51.1	50.1	56.0	61.4	69.0	72.8	75.1	73.6	71.6	66.6	57.0	51.2	(29)
(30)			MCDWB	54.1	54.0	61.8	70.3	77.5	80.5	82.5	80.1	78.1	70.6	61.1	53.6	(30)
(31)		5%	WB	44.6	44.6	52.3	58.6	66.8	71.1	73.2	71.7	69.5	63.0	53.9	46.1	(31)
(32)			MCDWB	47.3	48.7	57.6	66.9	74.5	78.9	79.9	77.9	75.3	67.5	58.2	48.4	(32)
(33)		10%	WB	38.1	38.3	46.9	55.3	64.4	69.2	71.5	70.3	67.5	59.5	50.3	41.9	(33)
(34)			MCDWB	41.2	42.5	52.6	62.9	71.0	76.1	77.3	76.0	73.1	64.5	55.4	45.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.4	13.8	15.4	18.6	18.5	17.1	17.0	16.5	16.7	15.1	13.2	10.7	(35)
(36)			MCDBR	18.0	19.8	21.9	24.2	21.2	19.3	18.9	18.5	19.8	19.5	18.5	16.5	(36)
(37)			MCWBR	15.9	16.4	14.5	14.7	11.2	10.5	9.0	9.2	9.8	11.4	13.3	14.7	(37)
(38)		5% WB	MCDBR	17.2	19.5	19.9	21.0	18.7	17.3	16.2	15.7	16.5	16.8	17.0	16.4	(38)
(39)			MCWBR	16.5	17.3	14.8	14.7	11.5	10.5	9.0	8.9	9.4	12.0	14.0	15.5	(39)
(40)		Clear-Sky Solar Irradiance	taub		0.297	0.327	0.342	0.371	0.408	0.433	0.432	0.420	0.375	0.352	0.310	0.293
(41)	taud		2.422	2.312	2.359	2.336	2.250	2.214	2.235	2.261	2.362	2.430	2.525	2.494	(41)	
(42)	Ebn at Noon		271	277	287	285	276	268	267	267	272	266	266	265	(42)	
(43)	Edh at Noon		25	32	35	38	43	45	43	41	35	29	23	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg		439	704	1022	1398	1666	1825	1843	1640	1315	835	539	337	(44)
(45)		RadStd		40	82	113	148	155	153	142	89	132	83	77	32	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+176	N/A

Nomenclature: See separate page